

Notes: Coate and Loury (AER, 1993)

Will Affirmative-Action Policies Eliminate Negative Stereotypes?

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1 One-sentence summary

This note summarizes Coate and Loury’s paper, which builds a statistical-discrimination model with endogenous worker investment to ask whether affirmative-action policies can eliminate negative stereotypes; the paper shows that equal-assignment affirmative action can sometimes break discriminatory beliefs, but can also preserve or worsen stereotypes through “patronizing” equilibria in which employers lower standards for the disadvantaged group, thereby reducing that group’s incentives to acquire skills.

2 Big picture

- **Question.** Can affirmative-action policies create temporary labor-market gains for minority workers without becoming a permanent fixture, or can they leave employer stereotypes intact?
- **Core mechanism.** Employers cannot perfectly observe workers’ productivity before job assignment. They observe group identity and a noisy signal. If they believe one group is less likely to be qualified, they interpret the same signal less favorably for that group.
- **Self-fulfilling stereotypes.** Workers decide whether to invest in becoming qualified before being matched with employers. Employer beliefs determine job-assignment standards, and these standards determine the expected return to investing. Negative beliefs can therefore reduce investment by the stereotyped group, validating the beliefs *ex post*.
- **Affirmative action in the model.** The policy is represented as an equal-assignment constraint: employers must assign workers from the two groups to the more rewarding task at the same rate.
- **Main tension.** Equal-assignment constraints can force employers to use group-conscious assignment rules. But the way they comply matters. If they lower standards for the disadvantaged group rather than revise beliefs, the policy can reduce investment incentives and sustain the stereotype.
- **Main result.** Affirmative action does not necessarily eliminate negative stereotypes. For an open set of parameters, a stable equilibrium under affirmative action continues to contain negative stereotypes about the initially disadvantaged group.
- **Policy implication.** Policies aimed directly at worker investment, especially subsidies to disadvantaged workers for acquiring productive skills, can be more reliable than policies that simply force assignment outcomes or subsidize employers.

3 Incremental contribution of the paper

3.1 A theory of affirmative action with endogenous stereotypes

The paper moves beyond the question of whether affirmative action raises minority representation in good jobs. It asks whether the policy changes the deeper equilibrium beliefs that generated unequal treatment in the first place. This matters because temporary improvements in job assignments may not persist if employer beliefs remain negative.

3.2 Linking statistical discrimination to worker investment incentives

Earlier statistical-discrimination models show that group identity can influence employer inference. Coate and Loury add an endogenous skill-acquisition decision. This transforms stereotypes into an equilibrium system: beliefs affect standards, standards affect investment, and investment outcomes validate or invalidate beliefs.

3.3 Distinguishing assignment equality from belief equality

The paper highlights a subtle distinction. A regulator may force equal assignment rates across groups, but this does not automatically imply that employers hold equal beliefs about the groups' productivity. Employers may satisfy the assignment rule through different standards, preserving unequal perceived productivity.

3.4 Introducing the patronizing-equilibrium logic

The paper's most distinctive contribution is the idea of a *patronizing equilibrium*: employers believe the disadvantaged group is less productive, comply with affirmative action by lowering that group's assignment standard, and thereby weaken that group's incentive to invest. The policy can then unintentionally confirm the original negative stereotype.

3.5 Policy ranking insight

The paper shows why different policy instruments with similar immediate targets can have very different dynamic consequences. Employer subsidies can backfire if they lower standards, while worker subsidies for acquiring skill can eliminate negative stereotypes if sufficiently large.

4 Model setup

4.1 Groups, jobs, and assignment

The economy contains many employers and a larger population of workers. Workers belong to one of two observable groups, denoted B and W . A fraction λ of workers are in group B , and the rest are in group W . Each employer assigns a worker to one of two jobs or tasks:

- **Task zero:** lower-return task.
- **Task one:** more demanding and more rewarding task.

Only qualified workers can perform task one satisfactorily. The employer wants to assign a worker to task one only if the worker is sufficiently likely to be qualified.

4.2 Worker investment

Workers are initially unqualified but can invest in becoming qualified. Investment is costly and heterogeneous across workers. A worker invests if the expected benefit of acquiring skill exceeds the cost. Thus the fraction of qualified workers in each group is endogenous.

4.3 Signals and employer beliefs

Employers observe a worker's group identity and a noisy signal about qualification. The signal is useful but imperfect. Let π_i denote the employer's prior belief that a worker from group $i \in \{B, W\}$

is qualified. A lower π_i means that the employer is more pessimistic about that group. Given the same signal, workers from the group with the lower prior are less likely to be assigned to the high-return task.

4.4 Assignment standard

Employers choose a standard s_i for each group. A worker from group i is assigned to task one if the observed signal is better than the chosen standard. More demanding standards reduce the risk of assigning unqualified workers to task one, but also reduce the probability that qualified workers get the high-return task.

4.5 Equilibrium structure

An equilibrium requires consistency between employers' beliefs and workers' actual investment decisions. If employers believe that group B is less likely to be qualified, they choose a stricter or otherwise less favorable assignment rule for B . Workers in group B then face lower expected returns to skill acquisition, invest less, and make the original belief correct in equilibrium.

5 Self-fulfilling negative stereotypes

5.1 How stereotypes arise

The paper's first major result is that discriminatory beliefs can arise even if the two groups are ex ante identical. The reason is strategic complementarity between employer beliefs and worker investment. If employers expect fewer B workers to be qualified, B workers receive weaker incentives to invest, causing fewer B workers to become qualified.

5.2 The role of the EE and WW curves

The paper uses a graphical fixed-point representation. The horizontal axis measures the assignment standard s , and the vertical axis measures the perceived fraction π of qualified workers. Two loci matter:

- **EE curve.** The employer-equilibrium curve maps beliefs into the optimal standard. It captures how employers assign workers given their belief about the group's productivity.
- **WW curve.** The worker-investment curve maps assignment standards into the realized fraction of workers who invest and become qualified.

An equilibrium occurs where the curves intersect. Multiple intersections mean multiple self-confirming belief systems.

5.3 Discriminatory equilibrium

A discriminatory equilibrium is one in which $\pi_B < \pi_W$. The two groups are objectively identical before the interaction, but because employers treat them differently, their skill-acquisition incentives differ. This generates different qualification rates in equilibrium.

5.4 Pareto ranking of equilibria

When there are multiple stable equilibria, the equilibrium with higher investment and less negative stereotyping can be Pareto superior. Employers benefit because they make better assignments, and workers benefit because the high-return task becomes more attainable. The paper therefore emphasizes that stereotypes are not just distributional; they can also reduce efficiency.

6 Affirmative action in the model

6.1 Equal-assignment constraint

The policy is modeled as a constraint that employers assign members of B and W to task one at the same rate. This is a natural representation of a quota-like or result-oriented affirmative-action policy. The regulator does not directly observe worker investment, productivity, or employer beliefs; it observes assignment outcomes.

6.2 Why the constraint can help

If the policy forces employers to treat the two groups symmetrically and if workers respond by investing at similar rates, then negative stereotypes can disappear. In such a case, affirmative action can be temporary: once beliefs become color-blind, the constraint may no longer be needed.

6.3 Why the constraint can fail

The constraint can also make employers use two different standards. If employers continue to believe that B workers are less productive, they may lower the standard for assigning B workers to task one while raising the standard for W workers. This preserves equal assignment rates but does not equalize beliefs. In fact, the lower standard for B may make investment less valuable, because even less-skilled B workers can obtain task one. This reduces the incentive of B workers to become qualified.

6.4 Patronizing equilibrium under affirmative action

The paper calls this outcome a patronizing equilibrium. Employers “help” the disadvantaged group in assignment, but in a way that confirms their pessimistic belief about the group’s productivity. In this equilibrium, affirmative action does not eliminate the stereotype; it embeds it into the assignment rule.

7 The main result and policy interpretation

7.1 Main theorem

The paper proves that for a nonempty open set of parameters, an affirmative-action equilibrium with negative stereotypes exists. This result is stronger than an example: it says that the failure of affirmative action to eliminate stereotypes is not knife-edge.

7.2 Economic intuition

The equal-assignment constraint creates a shadow price of equality. Employers must modify standards to comply with the constraint. If they initially believe that B workers are less productive,

the least costly way to satisfy equal assignment may be to assign more B workers through a lower standard rather than to revise beliefs. That change can reduce the return to skill acquisition for B workers.

7.3 When affirmative action succeeds

Affirmative action can succeed when the induced changes in assignment standards raise the returns to investment for the disadvantaged group and make discriminatory beliefs unstable. This is more likely when workers respond strongly to improved incentives and when employer beliefs are sufficiently responsive to evidence.

7.4 When affirmative action backfires

The policy can backfire when it patronizes the disadvantaged group. In that case, the assignment rate is equalized but the incentive to become qualified is weakened. The disadvantaged group may invest less, which reinforces the employers' belief that it is less productive.

7.5 Important distinction

The paper does not argue that affirmative action always fails. Instead, it identifies a precise mechanism through which affirmative action can fail. The implication is that policy design should focus not only on assignment outcomes, but also on investment incentives and the evolution of beliefs.

8 Further policy considerations

8.1 Changing employer preferences

A regulator could try to make employers prefer placing B workers in task one by altering payoffs. The paper shows that such interventions can be ambiguous. If the intervention effectively lowers the assignment standard for B without raising investment incentives, it can increase skill disparity.

8.2 Employer subsidies

A subsidy paid to employers for assigning B workers to task one can backfire. A marginal employer subsidy shifts the relevant assignment curve, but it can also lower standards for B . A large employer subsidy can make employers assign B workers even when they do not invest, producing an equilibrium with lower skill acquisition.

8.3 Worker subsidies

A subsidy paid directly to B workers for achieving task-one qualification has better properties. It raises the benefit of investment for disadvantaged workers and can eliminate negative stereotypes if it is large enough. The paper concludes that, generally speaking, it is better to subsidize disadvantaged workers for achieving good jobs than to subsidize employers for promoting them.

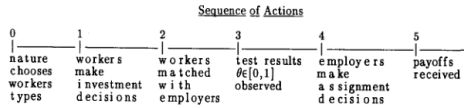
8.4 Policy lesson

The main policy lesson is that assignment rules alone may not solve the underlying equilibrium problem. A policy must alter the dynamic feedback between beliefs and investment. In this model, direct investment incentives are more reliable than policies that only regulate employer assignment behavior.

9 Tables and figures

The paper contains no empirical tables. Its core arguments are developed through nine figures that map beliefs, standards, worker investment incentives, and policy interventions. The figures below are extracted from the paper and placed in paired rows where possible.

9.1 Figure 1: Sequence of Actions and Figure 2: An Equilibrium with Negative Stereotypes Against B's



(a) **Figure 1: Sequence of Actions.** Timing of nature, worker investment, matching, employer signals, assignment, and payoffs.

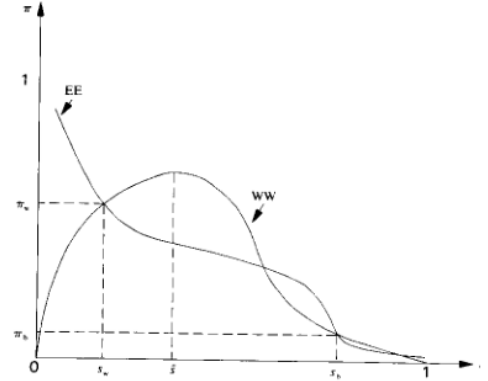


FIGURE 2. AN EQUILIBRIUM WITH NEGATIVE STEREOTYPES AGAINST B'S

(b) **Figure 2: An Equilibrium with Negative Stereotypes Against B's.** Intersection of employer standards and worker investment responses under discriminatory beliefs.

Figure 1 interpretation. The timing is central to the paper. Workers invest before they know the employer who will evaluate them, while employers set standards after observing group identity and a noisy signal. This sequencing creates the possibility of self-fulfilling beliefs: employers' expected treatment affects workers' investment decisions, and those investment decisions then validate or refute employer expectations.

Figure 2 interpretation. Figure 2 shows how a discriminatory equilibrium can arise even when the two groups are ex ante identical. The *EE* curve summarizes employer standards, while the *WW* curve summarizes worker investment. Multiple intersections imply multiple self-confirming equilibria. A low-belief equilibrium for group *B* can persist because stricter or less favorable assignment expectations reduce the return to investing.

9.2 Figure 3: Employers' Optimal Standards and W's Given Beliefs and Positive Value of Multiplier and Figure 4: An Equilibrium Under Affirmative Action with Negative Stereotype About B's

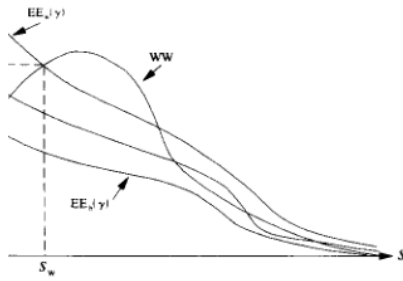


FIGURE 3. EMPLOYERS' OPTIMAL STANDARDS AND W'S GIVEN BELIEFS AND POSITIVE VALUE OF MULTIPLIER

(a) **Figure 3: Employers' Optimal Standards and W's Given Beliefs and Positive Value of Multiplier.** Affirmative action changes employers' constrained standard-setting problem.

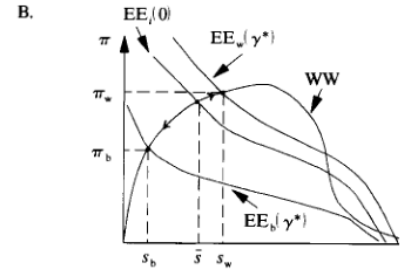
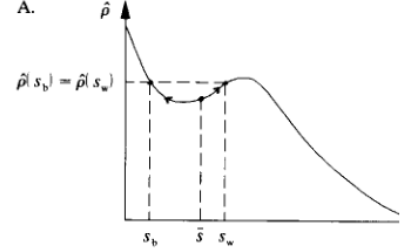


FIGURE 4. AN EQUILIBRIUM UNDER AFFIRMATIVE ACTION WITH NEGATIVE STEREOTYPE ABOUT B'S

(b) **Figure 4: An Equilibrium Under Affirmative Action with Negative Stereotype About B's.** Construction of an affirmative-action equilibrium that still contains negative stereotypes.

Figure 3 interpretation. The figure introduces the constrained assignment problem. With a positive multiplier on the affirmative-action constraint, employers' standards for the two groups move in different directions. The graph shows that equal assignment does not necessarily require equal beliefs. It can instead require group-specific standards that satisfy the quota constraint.

Figure 4 interpretation. Figure 4 is the graphical core of the paper's main result. It shows that under affirmative action, the system can still have an equilibrium with negative stereotypes against *B*. The policy changes the standards, but the induced standards can still be self-confirming. This illustrates why affirmative action can fail to eliminate the underlying discriminatory belief.

9.3 Figure 5: Affirmative Action Increases Skill Disparity in the Absence of Stereotypes and Figure 6: Marginal Employer Subsidy

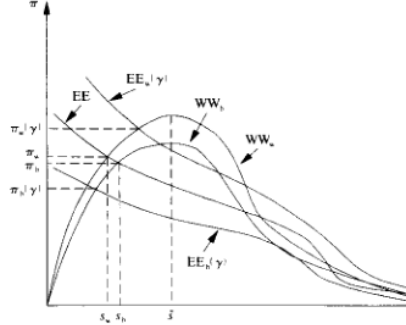


FIGURE 5. AFFIRMATIVE ACTION INCREASES SKILL DISPARITY IN THE ABSENCE OF STEREOTYPES

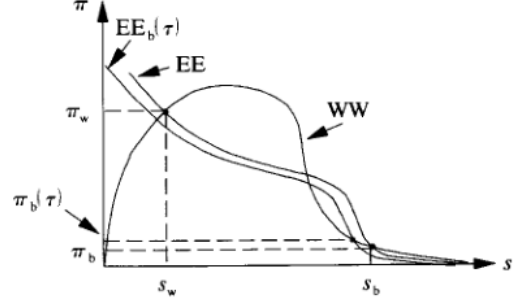


FIGURE 6. MARGINAL EMPLOYER SUBSIDY

(a) **Figure 5: Affirmative Action Increases Skill Disparity in the Absence of Stereotypes.** Targeting assignment preferences can create skill differences even when initial beliefs are color-blind.

(b) **Figure 6: Marginal Employer Subsidy.** A small subsidy to employers changes the assignment curves but may not eliminate negative stereotypes.

Figure 5 interpretation. Figure 5 warns that some forms of preferential assignment can create unintended skill disparities. Even without negative stereotypes, a policy that changes employer preferences toward group B can reduce the payoff to skill acquisition for B , thereby widening actual productivity differences. This is an important generalization of the patronizing logic.

Figure 6 interpretation. Figure 6 studies an employer subsidy that rewards the assignment of B workers to task one. A marginal subsidy may shift employer curves and alter beliefs, but it does not necessarily break the negative stereotype. The result depends on whether the subsidy induces a self-reinforcing movement toward more investment or merely changes assignment standards.

9.4 Figure 7: Marginal Worker Subsidy and Figure 8: Large Employer Subsidy

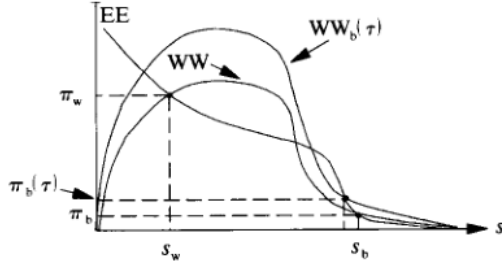


FIGURE 7. MARGINAL WORKER SUBSIDY

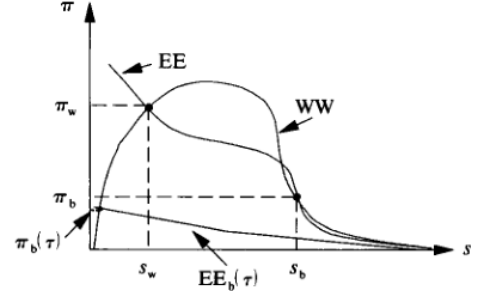


FIGURE 8. LARGE EMPLOYER SUBSIDY

(a) **Figure 7: Marginal Worker Subsidy.** A small direct subsidy to workers can change worker incentives more directly than employer subsidies.

(b) **Figure 8: Large Employer Subsidy.** A large employer subsidy can backfire by inducing lower standards for B workers.

Figure 7 interpretation. Figure 7 shows a worker subsidy that raises the return to investment for group B. Unlike an employer subsidy, it directly changes workers' incentives to become qualified. If the subsidy moves the system out of the basin of attraction of the negative-stereotype equilibrium, it can help eliminate discriminatory beliefs.

Figure 8 interpretation. Figure 8 shows why subsidizing employers can be dangerous. A large employer subsidy can lead employers to lower standards for B workers so much that investment becomes less valuable. The policy then reinforces, rather than eliminates, negative beliefs.

9.5 Figure 9: Large Worker Subsidy

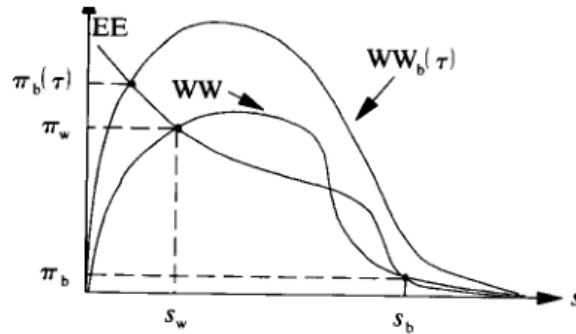


FIGURE 9. LARGE WORKER SUBSIDY

Figure 9: Large Worker Subsidy. A large direct subsidy to B workers can eliminate the negative-stereotype equilibrium.

Figure 9 interpretation. Figure 9 is the optimistic counterpart to Figure 8. A sufficiently large worker subsidy raises the payoff to acquiring skills enough to eliminate all locally stable negative-

stereotype equilibria. The economy can then transition toward a nondiscriminatory, Pareto-superior equilibrium. This is why the paper concludes that direct worker investment incentives are generally more reliable than employer-side incentives.

10 Short summary

- The paper asks whether affirmative action can eliminate negative stereotypes rather than merely improve assignment outcomes.
- The model contains two groups, noisy signals, employer assignment standards, and worker skill-investment decisions.
- Negative stereotypes are self-fulfilling when pessimistic employer beliefs reduce the stereotyped group's expected return to investment.
- Affirmative action is modeled as an equal-assignment constraint requiring the two groups to be assigned to the desirable task at equal rates.
- Equal assignment can eliminate stereotypes in some circumstances, but it need not do so.
- A patronizing equilibrium occurs when employers comply by lowering standards for the disadvantaged group while maintaining pessimistic beliefs about that group's productivity.
- In a patronizing equilibrium, affirmative action may reduce the disadvantaged group's incentives to invest, causing the stereotype to persist or worsen.
- The main theoretical result shows that such equilibria exist for an open set of parameters.
- Employer subsidies can backfire if they lower standards without improving investment incentives.
- Worker subsidies targeted at skill acquisition are more likely to eliminate negative stereotypes because they directly raise the return to becoming qualified.

11 Conclusion

11.1 Core conclusion

Coate and Loury show that affirmative action does not mechanically eliminate negative stereotypes. Even if a policy equalizes assignment rates, employers may continue to hold pessimistic beliefs and may satisfy the policy by changing standards in a way that weakens worker investment incentives. Therefore, affirmative action can sometimes be temporary and successful, but it can also become permanent because it fails to change the underlying belief-investment feedback loop.

11.2 Economic intuition

The economic intuition is that stereotypes are equilibrium objects. They are not simply incorrect ideas that disappear once workers receive opportunities. Employers' beliefs shape the returns to acquiring skills. If those beliefs lead to unfavorable assignment rules, workers rationally invest less. If affirmative action changes assignment outcomes without raising the return to skill acquisition, it may not solve the problem.

11.3 Incremental contribution revisited

The paper's contribution is to embed affirmative action in a statistical-discrimination model with endogenous human-capital investment. This framework separates representation, beliefs, and skill acquisition. It shows that equalizing one dimension - assignments - may leave the other dimensions unchanged or worse.

11.4 Policy implication

The paper suggests that policy should be evaluated by its effect on investment incentives and belief dynamics, not only by its effect on immediate job assignment rates. Policies that help disadvantaged workers acquire demonstrable skills may be more effective than policies that merely induce employers to assign more workers from the disadvantaged group to desirable jobs.

11.5 Limitations

The model is theoretical and stylized. It abstracts from many institutional details, such as legal enforcement, wage bargaining, multi-period careers, networks, firm heterogeneity, and richer forms of human capital. It also treats group membership and signals in a simplified way. These abstractions are useful for isolating the mechanism, but empirical validation requires separate evidence.

11.6 Takeaway

The enduring takeaway is that affirmative action must be designed as a dynamic policy. If it changes only assignment rates, it may leave self-fulfilling stereotypes intact. If it changes the incentives that generate skill investment and belief revision, it has a stronger chance of producing a lasting color-blind outcome.